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RESEARCH ARTICLE

A Dynamic Landslide Warning Model Based on Grey System Theory

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ABSTRACT Landslides are among the most frequent and destructive geological disasters, causing significant loss of life and property. Research on landslide prediction is crucial for minimizing disaster-related losses, ensuring public safety, and developing reliable early warning systems and preparedness strategies. Existing prediction technologies are broadly classified into mechanical modeling methods and mathematical modeling methods. Compared to mechanical modeling, mathematical modeling relies on data analysis, enabling the use of existing monitoring data without requiring detailed geological information. Among these, the grey model stands out for its high accuracy with minimal information requirements, as well as its simplicity and ease of implementation. Based on these advantages, grey model-based landslide prediction has emerged as a key research focus. This paper proposes a dynamic landslide warning model that integrates the concept of risk with grey system theory, using a small amount of real-time monitoring data. Experimental results demonstrate the model's effectiveness in accurately predicting landslides and providing timely warnings for unstable slopes.

INDEX TERMS Grey model, sliding window, dynamic prediction, landslide warning model.

I. INTRODUCTION

Landslides are common and sudden geological disasters that can cause significant damage and are challenging to manage or mitigate. Without accurate early warning systems and proper emergency response measures, landslides can result in substantial loss of life and property. Every year, landslides lead to thousands of casualties. Developing effective warning systems is critical to reducing disaster-related losses and saving lives, making it an important area of research.

Landslide prediction methods can be broadly classified into mechanical and mathematical modeling approaches. Mechanical models offer several advantages, such as a clear understanding of physical mechanisms, wide applicability, and the ability to capture complex mechanical behaviors like stress distribution and friction. These features provide valuable insights into the landslide process. However, mechanical models are often complex and require detailed geological data, which can be difficult to obtain. Additionally, uncertainties in the model parameters often require manual adjustments to ensure an accurate fit.

Extensive landslide data shows that slope deformation progresses through three distinct phases under the influence of gravity, as shown in Figure 1. In the initial deformation phase (AB section), the slope begins to deform, and the deformation curve is steep, indicating rapid structural changes. As time passes, the deformation rate slows, leading to the next phase. The uniform deformation phase (BC section) is characterized by a more consistent deformation rate, with gravitational forces continuing to influence the slope. Though fluctuations may occur, the deformation curve can be approximated by a straight line with a slight tilt. In the accelerated deformation phase (CD section), the deformation rate increases significantly, causing rapid structural changes

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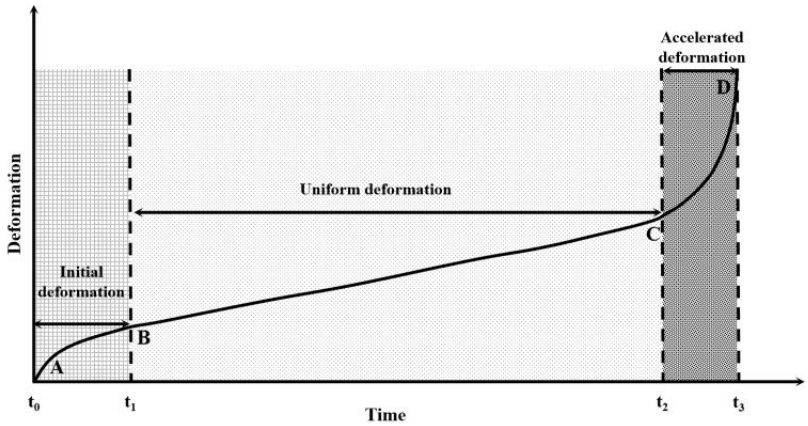


FIGURE 1. Deformation evolution curve of slope.

in the slope. The deformation curve steepens, ultimately leading to a landslide as the process reaches its peak.

The evolution of slope deformation is closely related to grey model predictions. For the slope deformation shown in Figure 1, when monitoring data is selected appropriately, the grey model predictions in the AB and BC sections align well with the actual monitoring values. However, in the CD section, the predictions tend to be lower than the actual values. Based on this observation, this study employs a grey model to predict long-term slope deformation. A risk model is then developed by analyzing the discrepancy between the grey model predictions and the actual monitoring data. This risk model is used to issue a three-level warning for the potential landslide.

The remainder of this paper is organized as follows: Section II reviews the current research on landslide prediction and early warning systems, including methods based on grey models and other mathematical approaches. Section III presents the dynamic landslide warning model proposed in this study. Section IV validates the proposed model by conducting experiments using the Preonzo slope public datasets. Finally, Section V discusses the proposed model and future research directions, Section VI concludes the paper.

II. RELATED WORKS

Grey System Theory, first proposed by Deng in 1982 [1], is particularly useful for making accurate predictions using small-scale data sequences. This approach employs differential equations to analyze the underlying patterns within data, requiring minimal information for modeling. Grey models have been widely applied in predicting various factors such as population growth [2], grain output [3], GDP [4], and energy consumption evaluation [5]. In the field of landslide prediction, several methods based on grey models have also been developed.

However, Traditional grey model is primarily suited for short-term predictions and is difficult to apply effectively for long-term forecasting. To address the long-term prediction of landslides, Zhang et al. enhanced the traditional GM(1,1) grey model by incorporating nonlinear time correction factors, which improved its accuracy for long-term deformation predictions. In their work, they modified the traditional GM(1,1) model into the TGM(1,1,p,q) model by adding nonlinear time correction factors and optimizing the background values using Genetic Algorithms, yielding good experimental results on the Majiagou slope [6]. While grey system theory is effective for linear problems, landslide prediction is a nonlinear problem over the long term. To enhance the accuracy of nonlinear landslide predictions, Li and Wu developed a grey model by combining Twin Support Vector Regression with a Hausdorff Derivative Operator. This model is a non-linear data-driven approach, which uses an improved salp swarm algorithm to determine the model's parameters [7]. To address the shortcomings of neglecting time factors, Wu et al. incorporated a time factor into the grey model and proposed a background value optimization nonlinear grey model (BNGM(1,1,t^2)) for landslide prediction, which performed well in four typical landslide cases [8]. The selection of prediction models and their practical applicability are crucial factors in slope deformation prediction. To meet these requirements, Li et al. proposed a fractional-order calculus grey model that couples grey theory with the fractional derivative method. This model minimizes added complexity while maintaining a reasonable level of accuracy [9]. To further improve predictive accuracy, combining grey models with other models has proven to be an effective approach. Li et al. integrated score modeling, kernel function methods, and grey models to propose a Kernel Grey Model with Fractional Operators (FKGM), which effectively predicted landslides in Hubei, China [10]. Miao et al. combined the Saito landslide prediction method with a time prediction model and the Verhulst grey model to improve prediction accuracy. Their integrated model was successfully tested on both mining and natural slope data [11]. In addition to the Verhulst model and GM(1,1), Liu et al. integrated the Saito model with machine learning techniques to assess

the contribution of each model, further improving prediction accuracy [12].

In addition to grey models, various mathematical models are also widely used for landslide prediction, including artificial neural networks (ANN) [13], [14] [15], support vector machines (SVM) [16], [17], genetic algorithms (GA) [18], [19], decision trees [20], [21], and random forests [22], [23]. These methods generally require initial data for training, and the accuracy of predictions depends on the quality of the training data. To improve prediction accuracy, large datasets are necessary. However, due to challenges in data acquisition, it is difficult to obtain large-scale, high-dimensional real-time data, which complicates dynamic, real-time predictions for different scenarios. Given the complexities and precision requirements of large-scale real-time monitoring data, using small-scale monitoring data for landslide predictions is often more feasible in practical engineering applications. For instance, grey models are often applied for predictions in such cases.

In scenarios with limited sample data, SVM typically shows better performance and is widely used in slope displacement prediction. However, SVM is highly sensitive to parameter selection. To address this, Zhou et al. proposed a particle swarm optimization and support vector machine (PSO-SVM) coupling model based on support factor responses, addressing periodic slope displacement by combining PSO, SVM, and wavelet transformations [24]. They further developed the PSO-Kernel Extreme Learning Machine (PSO-KELM) method to improve prediction accuracy in these scenarios [25]. The prediction of potential landslides based on historical records has also become a significant research focus. Rossi et al. used the Zipf distribution to model landslide risk, based on data from 958 locations in Italy between 1861 and 2015, effectively predicting the frequency and probability of fatal landslides [26].

In recent years, neural networks have also been widely used in landslide prediction. To reduce human bias and improve the accuracy of artificial neural network (ANN) predictions, Lian et al. introduced several switching prediction methods, emphasizing the importance of selecting the appropriate neural network model [27]. Shihabudhen et al. proposed an adaptive neural fuzzy reasoning system for predicting extreme landslide displacements, applying the model to the White Shuihe and ShiliuShubao landslides [28]. Meng et al. introduced a deep learning framework for landslide prediction that combines the Laplace Function Sparse Regularized Continuous Deep Belief Network (LSCDBN), Grey Wolf Optimization Algorithm (GWO), and Whale Optimization Algorithm (WOA). Using historical landslide data from Denmark, they effectively predicted subsequent landslides [29].

Traditional grey models can predict landslides relatively easily; however, their application for long-term nonlinear predictions is limited, affecting the accuracy of long-term forecasting. Machine learning-based methods can effectively predict long-term nonlinear slope displacement, but they require high-quality data and substantial amounts of training data, which makes their widespread application for displacement prediction and landslide early warning across various slopes challenging. In this paper, we combine the GM(1,1) model with the concept of a sliding window based on grey system theory. This ensures that only the most recent data is used for predictions, minimizing errors caused by outdated historical data. Additionally, a three-level early warning mechanism is established based on the prediction results and real-time monitoring data, and a landslide early warning model is proposed.

III. METHODOLOGY

In this section, a dynamic landslide warning model is proposed by integrating the GM (1,1) model with sliding window technology. The GM (1,1) model, a core component of grey system theory, is widely used for time series forecasting. Sliding window technology helps to process sequential data in a more efficient manner, allowing real-time analysis and updates to be achieved. By combining these two methodologies, the proposed warning model aims to provide more accurate and timely alerts for potential landslides.

A. DYNAMIC GREY MODEL FOR LANDSLIDE PREDICTION

The preprocessed equal time interval slope displacement data sequence and it's first-order accumulated generating operation sequence are defined as:

$X^{(0)} = (x^{(0)}(1),\ x^{(0)}(2),\ \ldots\ x^{(0)}(n))$ and
$X^{(1)} = (x^{(1)}(1),\ x^{(1)}(2),\ \ldots x^{(1)}(n))$ respectively,
where $x^{(1)}(k) = \sum_{i=1}^{k} x^{(0)}(i),\ k = 1, 2, \ldots, n$
Let $Z^{(1)} = (z^{(1)}(2), z^{(1)}(3), \ldots, z^{(1)}(n))$,
where $z^{(1)}(k) = (x^{(1)}(k) - x^{(1)}(k-1))/2$. Then, let

$$x^{(0)}(k) + az^{(1)}(k) = b \tag{1}$$

Equation (1) represents the mean form of the GM(1,1) model, which is fundamentally a difference equation. In this equation, -a denotes the development coefficient, and b represents the grey action quantity. The parameter vector $\hat{a} = [a, b]^T$ can be estimated using the least squares method.

$$\hat{a} = (B^T B)^{-1} B^T Y \tag{2}$$

where

$$Y = \begin{bmatrix} x^{(0)}(2) \\ x^{(0)}(3) \\ \cdots \\ x^{(0)}(n) \end{bmatrix},\quad B = \begin{bmatrix} -z^{(1)}(2) & 1 \\ -z^{(1)}(3) & 1 \\ \cdots & \cdots \\ -z^{(1)}(n) & 1 \end{bmatrix} \tag{3}$$

The cumulative reduction restoration formula of the GM(1,1) model is:

$$\hat{x}^{(0)}(k) = a^{(1)}\hat{x}^{(1)}(k) = \hat{x}^{(1)}(k) - \hat{x}^{(1)}(k-1),\ k = 2, 3, \ldots, n \tag{4}$$

Among them, the predictive value of the k-th element in the sequence of the slope displacement $\hat{x}^{(1)}(k)$ can be calculated

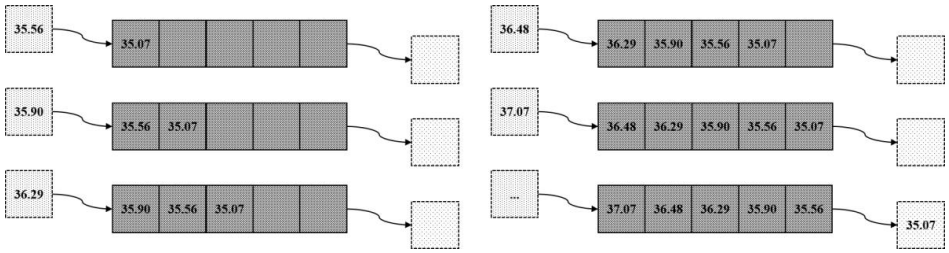


FIGURE 2. An example of sliding window.

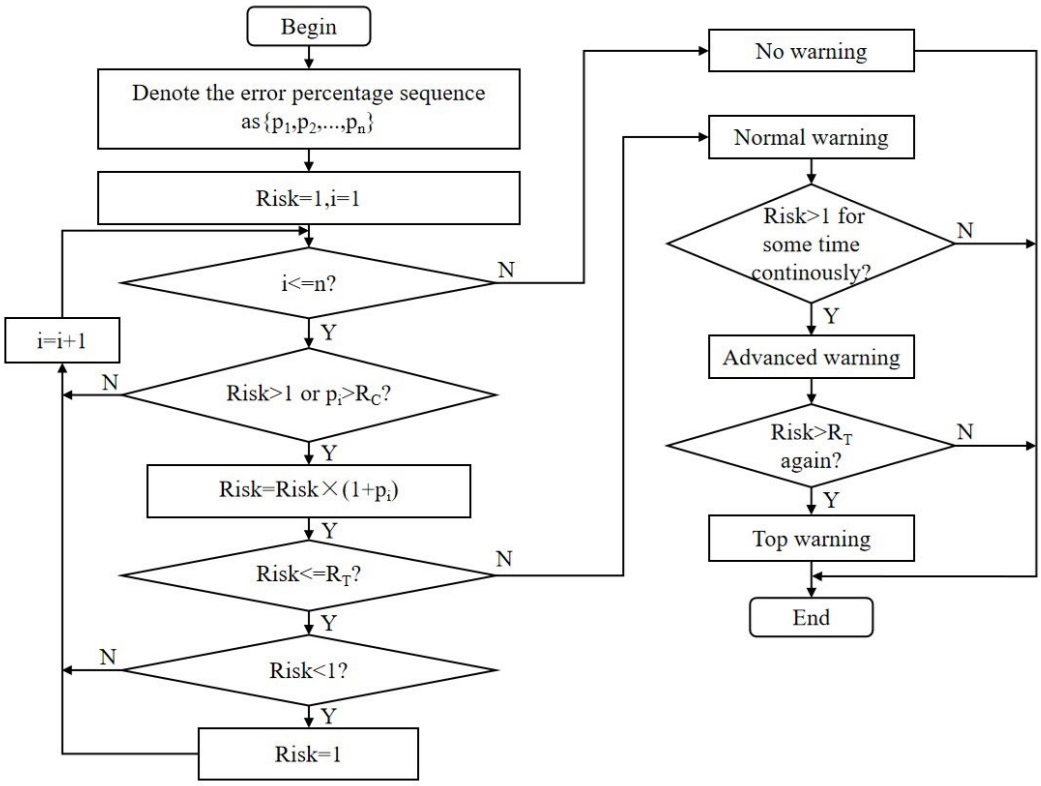


FIGURE 3. Flowchart of early warning.

based on a, b, and the first element $x^{(0)}(1)$ in the sequence.

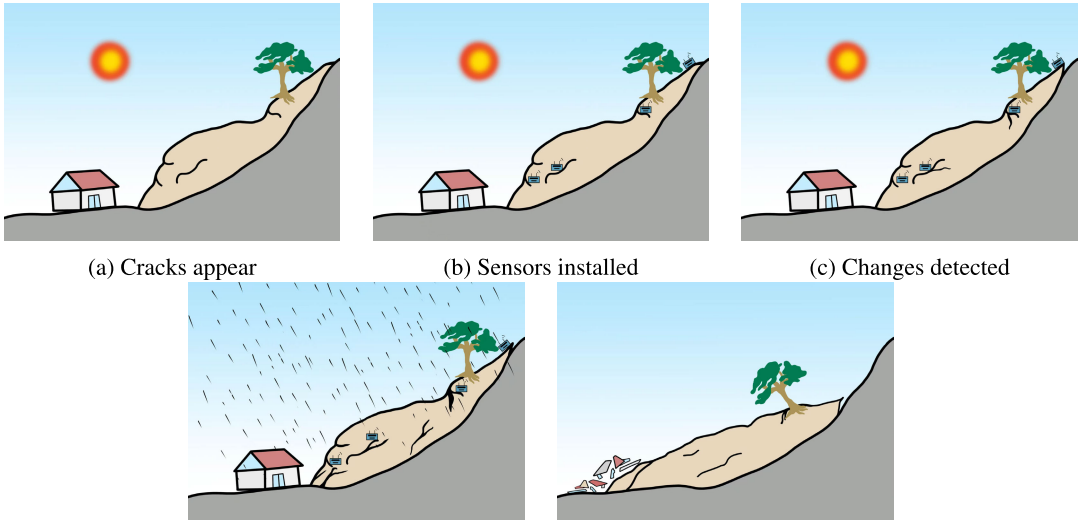
$$\hat{x}^{(1)}(k) = (x^{(0)}(1) - \frac{b}{a}) \times (\frac{1}{1+a})^{k-1} + \frac{b}{a}, k = 1, 2, 3 \ldots \tag{5}$$

Once the landslide monitoring data has been preprocessed and organized into equal time intervals, a GM (1,1) model can be established, and the development coefficient -a and grey action quantity b can be calculated.

Then, based on the cumulative reduction restoration formula (Eq. 4), $\hat{x}^{(1)}(n)$, $\hat{x}^{(1)}(n+1)$, $\hat{x}^{(1)}(n+2)$,...can be calculated. After that, the predictive slope displacement value $\hat{x}^{(0)}(n)$, $\hat{x}^{(0)}(n+1)$, $\hat{x}^{(0)}(n+2)$,...can be calculated based on the corresponding k.

Directly incorporating all available data, including newly acquired monitoring data, into a GM (1,1) model can potentially reduce the accuracy of predictions. This is because older data may contain irrelevant or outdated information that does not accurately represent the current state of the system. To address this issue, sliding window technology is employed to ensure that only the most recent and relevant data is considered. This method allows for a continuous updating process, where the most current information is used for prediction while older data is disregarded.

Figure 2 shows an example of a sliding window with a window length of 5 in the application of slope displacement prediction and landslide warning. Initially, the window only has the first displacement data (35.07mm). When the second data (35.56mm) is obtained, it is added after the first data point since the window is not yet full. As more data points are obtained, they are added to the window sequentially until it reaches its maximum capacity. Finally, as the window



(a) Cracks appear (b) Sensors installed (c) Changes detected

(d) Early warning required (e) Landslide occurs

FIGURE 4. Landslide and early warning.

reaches its maximum capacity of 5 data points, the addition of new data points, like the sixth data point (37.07mm), prompts the removal of the oldest data point (35.07mm) from the window, ensuring that only the most recent and relevant information is considered for prediction purposes. Sliding window technology effectively replaces old data with new, real-time monitored data. Using the latest information to build a grey model can make more accurate landslide predictions.

B. DYNAMIC LANDSLIDE EARLY WARNING MODEL

The error percentage between the predicted and actual displacements can be calculated by comparing the grey prediction model's forecasted displacement data with the actual monitoring data at the next time interval. Utilizing this information, a stability prediction algorithm is established. When the algorithm indicates a loss in stability, an early warning is issued, allowing for timely preventive measures and response strategies to be implemented in anticipation of potential landslide events.

The modified error of the time (Δ) corresponding to the *n-th* position in the sequence and the final displacement predicted value corresponding to the next monitoring time $y(n+1)$ can be defined as Eq. 6 and 7, respectively. As a result, the error percentage (p) between actual displacement and predictive displacement can be calculated by Eq. 8.

$$\Delta = \hat{x}^{(0)}(n) - x^{(0)}(n) \tag{6}$$

$$y(n+1) = \hat{x}^{(0)}(n+1) - \Delta \tag{7}$$

$$\mathrm{p} = \begin{cases} \dfrac{x^{(0)}(n+1) - \mathrm{y}(\mathrm{n}+1)}{\mathrm{y}(\mathrm{n}+1) - \mathrm{y}(n)} \times 100\%, & \mathrm{y}(\mathrm{n}+1) > y(n) \\ 0, & \text{Others} \end{cases} \tag{8}$$

Define $S_p = \{p_1, p_2, \ldots, p_n\}$ as the sequence of error percentages calculated from the differences between predicted and actual displacements. This sequence is then used to predict the slope's stability and determine the need for warnings.

This paper divides landslide warnings into three distinct categories, each corresponding to a different level of urgency and required response.

The first category is the **Normal Warning**, which signifies a potential landslide occurrence on the slope. Under this warning, continuous monitoring and prediction are required to keep track of any changes in the slope's stability. If inconsistencies between predicted and actual displacements persist, the warning level will be escalated to Advanced Warning. However, if subsequent predictions align with monitoring results, the warning will be canceled, indicating that the slope has stabilized.

The second category is the **Advanced Warning**, which indicates a high likelihood of a significant landslide event occurring. In such cases, prompt action must be taken, which may include evacuation or other measures appropriate to the specific situation.

The third category is the **Highest Warning**, which implies an imminent landslide event. This level demands immediate action to ensure safety and mitigate potential damage.

Figure 3 illustrates the flowchart of early warning, and Algorithm 1 shows the corresponding calculation process:

The early warning threshold R_T serves as a crucial parameter in determining whether a warning should be issued. This threshold value can be manually specified based on the specific requirements of various applications. While it can be set to 4 (or 400%), it is generally recommended not to be lower than 3 (or 300%), which corresponds to a minimum acceptable margin of error between predicted and actual

Algorithm 1 Calculation Process of Early Warning

Require: $\{p_1, p_2, \ldots, p-n\}$, early warning threshold R_T, error calculation threshold R_C

Ensure: Whether to warn (or the level of warning)

1: $Risk = 1$
2: **for** each element p_i in sequence $\{p_1, p_2, \ldots, p_n\}$ **do**
3: **if** $Risk > 1$ or $p_i > R_C$ **then**
4: $Risk = Risk \times (1 + p_i)$
5: **end if**
6: **if** $Risk > R_T$ **then**
7: Output "Normal warning"
8: **if** in the follow-up for a period of time, $Risk > 1$ appears continuously **then**
9: Output "Advanced warning"
10: **if** $Risk > R_T$ appears again **then**
11: Output "Highest warning"
12: **end if**
13: **end if**
14: **else if** $Risk < 1$ **then**
15: $Risk = 1$
16: Output "No warning"
17: **end if**
18: **end for**

displacements. Steps 6-13 outline the process for issuing warnings when the *Risk* value exceeds the predetermined threshold R_T. Initially, when *Risk* becomes greater than R_T, a "normal warning" is issued. If *Risk* continues to remain above 1 for a specified duration, the warning level is upgraded to an "advanced warning". The length of this duration is linked to the size of the sliding window; for instance, if the sliding window covers 5 days, the specified duration could be set to 5 days. In cases where an "advanced warning" has already been issued and the real-time Risk value exceeds R_T once again, a "highest warning" should be issued. The error calculation threshold R_C is used to judge whether to use p_i to calculate *Risk*. R_C could be set as 50% or other values, but it is not recommended to be more than 100%. The selection of an appropriate error calculation threshold is crucial in identifying landslide trends and ensuring accurate predictions. If the threshold is set too high, there is a risk of overlooking significant changes in slope stability, which may lead to undetected landslides. On the other hand, setting the threshold too low can lead to unnecessary calculations and false alarms, creating inefficiencies and undermining the credibility of the warning system.

The generation of the error percentage sequence S_p involves a multi-step process to capture the evolution of discrepancies between predicted and actual displacement values.

First, the error percentage, p_i, is calculated using the predicted displacement value and the real-time displacement monitoring value. If p_i is greater than the critical error value R_C, it is recorded for further analysis.

Next, moving backward from p_i, error percentages are calculated for preceding data points p_{i-1}, p_{i-2}, and so on, as long as p_k $(k = i-1,\ i-2,\ i-3, \ldots)$ remains greater than 0.

Finally, as new displacement monitoring data becomes available in real-time, the error percentages p_{i+1}, p_{i+2}, p_{i+3}, etc. are calculated and recorded. This process continues until the early warning process is complete.

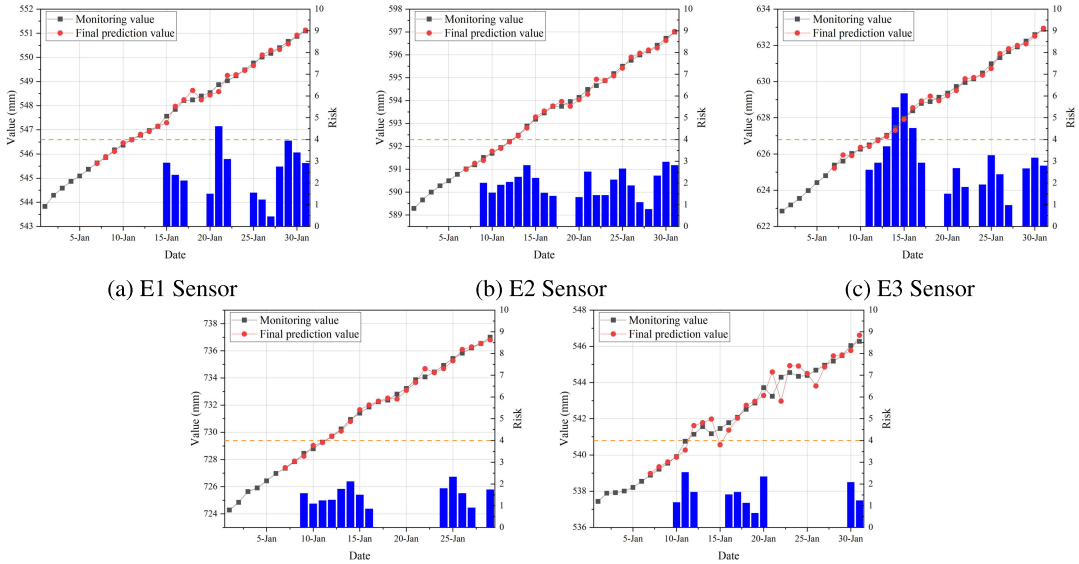
Figure 4 depicts the progression of a landslide alongside a schematic representation of monitoring and early warning mechanisms. In Figure 4(a), cracks begin to form at the rear edge of the slope, necessitating close monitoring. In Figure 4(b), sensors are deployed at the rear edge and other cracked areas to enable real-time data collection. In Figure 4(c), localized creeping of the slope becomes evident, with sensors continuing to monitor relevant data. In Figure 4(d), external factors such as heavy rainfall cause a significant discrepancy between the sensor readings and predicted values, signaling an imminent landslide and triggering an urgent early warning. Finally, in Figure 4(e), the landslide occurs. Without timely warnings and appropriate evacuation measures, such an event could lead to substantial losses. The early warning model proposed in this paper will issue different levels of warnings during the stages illustrated in Figures 4(c)-(d).

IV. EXPERIMENTS

To demonstrate the differences in performance between the slope stability phase and the pre-landslide phase, we use the Preonzo slope dataset [30] to evaluate the dynamic prediction model (Sections IV-A and IV-B). This dataset includes displacement data collected from five extensometer sensors on the Preonzo slope between 2002 and 2012. Two specific periods are analyzed: the stable phase (January 1 to January 31, 2012) and the pre-landslide phase (April 10 to May 15, 2012).

To further evaluate the model's applicability in various scenarios, two different landslide datasets: the Wolongsi landslide and the Longjingcun landslide datasets from Reference [12] are used to examine the performance in Section IV-C. These datasets help validate the effectiveness of the proposed early warning approach.

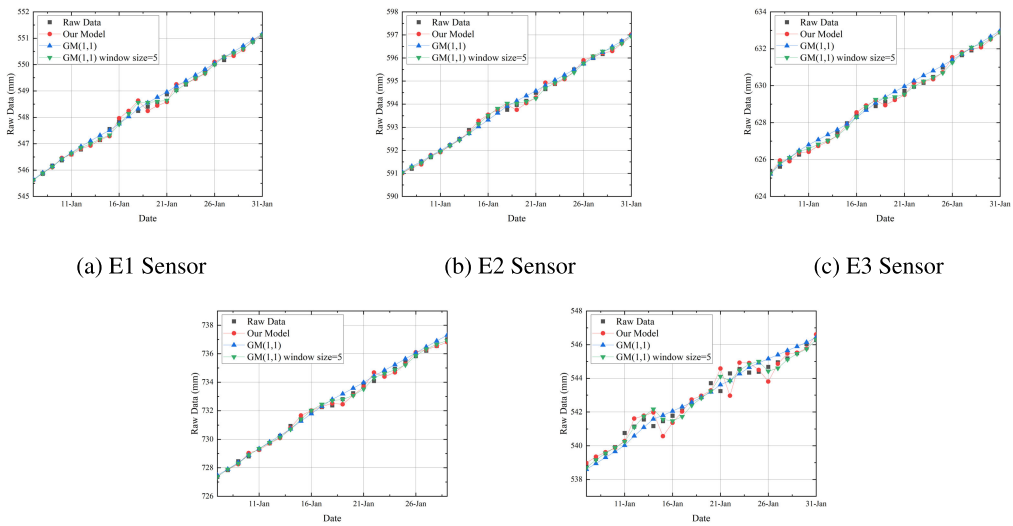
The analysis utilizes a warning threshold R_T of 400% and an error calculation threshold R_C of 50%. By applying these parameters, the study aims to assess the effectiveness of the dynamic prediction model in detecting changes in slope stability and providing timely warnings under different conditions. In the experiments, the process of real-time data acquisition was simulated, and prediction results were generated. For example, assuming "today" is April 30th, we used the April 30th data from the dataset to simulate the real-time data that would be monitored by the sensor on that day, and predicted the landslide risk for that day. The experimental results demonstrate that the proposed model is capable of issuing effective early warnings.



(a) E1 Sensor (b) E2 Sensor (c) E3 Sensor

(d) E4 Sensor (e) E5 Sensor

FIGURE 5. Experimental results of sensors when Preonzo slope was stable.



(a) E1 Sensor (b) E2 Sensor (c) E3 Sensor

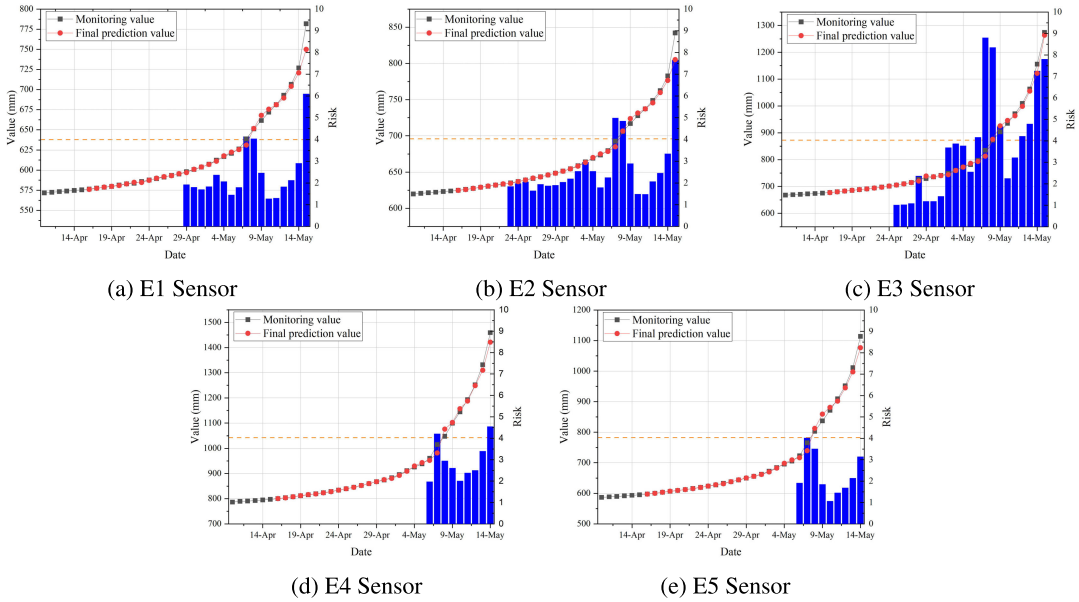
(d) E4 Sensor (e) E5 Sensor

FIGURE 6. Comparative results of different models when Preonzo slope was stable.

A. EXPERIMENTS ON STABLE SLOPE

This section focuses on the data collected from the five extensometer sensors (E1-E5) during the stable period of the Preonzo slope in January 2012. Figure 5(a)-(e) presents the experimental results obtained from each of these sensors throughout this time frame, and Table 1 illustrates the risk values corresponding to each sensor at that stage. The results illustrate the performance and behavior of the sensors when monitoring the slope under stable conditions.

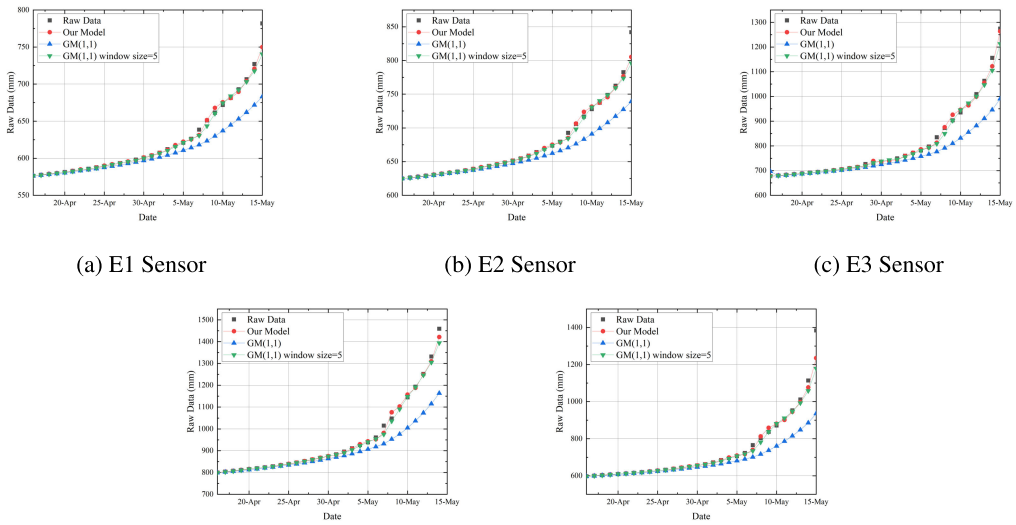
The results from the stable period in January 2012 reveal that the predicted slope displacement values obtained using the proposed model closely align with the actual displacement measurements. This close agreement between the predicted and actual values indicates that the model effectively captures the behavior of the slope under stable conditions and produces reliable predictions. Although the slope remained stable without any landslide occurrence during the analyzed period, two normal warnings were issued by the E1 sensor on January 21 and the E3 sensor on January 14 (the risk value exceeds 400%). The remaining sensors did not generate any early warnings, and no advanced warnings were issued by any of the sensors.



(a) E1 Sensor (b) E2 Sensor (c) E3 Sensor

(d) E4 Sensor (e) E5 Sensor

FIGURE 7. Experimental results of sensors before the landslide occurred on the Preonzo Slope.



(a) E1 Sensor (b) E2 Sensor (c) E3 Sensor

(d) E4 Sensor (e) E5 Sensor

FIGURE 8. Comparative results of different models before the landslide occurred on the Preonzo Slope.

Despite these two instances of normal warnings, the overall prediction results were largely consistent with expectations. The normal warnings were also relieved on January 23 and January 18, respectively. This suggests that while the model may occasionally generate normal warning under stable conditions, it generally performs well in monitoring slope stability and providing accurate predictions. The sliding window technique is employed to dynamically select the most recent data for prediction and early warning. Even if a small portion of the data is inconsistent or inaccurate, as long as the errors are not persistent, they do not affect the final warning outcome.

To assess the accuracy of the prediction model proposed in this study, we conducted comparative experiments involving the proposed model, the standard GM(1,1) model, and the GM(1,1) model with a window size of 5. Figure 6 illustrates the comparative results for the Preonzo slope during its stable phase (January 2012). As shown in the figure, during the slope's stable phase, the displacement values predicted by all three models closely aligned with the actual displacement values.

However, Table 2 summarizes the T-test results for the three prediction models. Paired-sample T-tests were performed on data collected between January 7 and

TABLE 1. Risk value of sensors when Preonzo Slope was stable.

Date	E1	E2	E3	E4	E5
9-Jan	-	200.00%	-	-156.76%	-
10-Jan	-	155.00%	-	109.14%	115.38%
11-Jan	-	190.77%	260.00%	124.02%	253.84%
12-Jan	-	204.40%	292.50%	126.77%	162.91%
13-Jan	-	227.98%	368.80%	176.82%	-
14-Jan	-	281.62%	**547.62%**	211.68%	-
15-Jan	292.86%	222.95%	611.51%	150.15%	-
16-Jan	236.87%	154.35%	451.57%	85.80%	151.25%
17-Jan	210.55%	140.32%	292.91%	-	162.71%
18-Jan	-	-	-	-	112.29%
19-Jan	-	-	-	-	66.35%
20-Jan	150.00%	134.48%	150.00%	-	234.38%
21-Jan	**460.71%**	252.15%	267.86%	-	-
22-Jan	309.43%	143.53%	181.32%	-	-
23-Jan	-	143.53%	-	-	-
24-Jan	-	215.30%	192.31%	180.00%	-
25-Jan	155.00%	265.96%	327.45%	232.76%	-
26-Jan	123.30%	188.39%	239.60%	157.04%	-
27-Jan	45.42%	110.81%	97.61%	90.92%	-
28-Jan	275.00%	79.15%	-	-	-
29-Jan	394.57%	233.33%	266.67%	174.07%	-
30-Jan	339.76%	296.97%	316.28%	-	208.00%
31-Jan	291.23%	281.74%	279.51%	-	123.81%

TABLE 2. Results of T-test on different models (when Preonzo Slope was stable).

Monitoring Point	Our Model	Grey Model with window size =5	Grey Model
E1 (DF=24)	-0.01412	-0.23482	3.87866
E2 (DF=24)	0.16798	1.49913	-1.70737
E3 (DF=24)	0.03529	0.39039	-4.98785
E4 (DF=23)	0.05823	-0.33324	-4.13357
E5 (DF=24)	-0.11483	0.15342	-0.30807

January 31 for each model. The results reveal that the proposed model achieved the smallest t-value, indicating the least difference between its predictions and the actual monitoring results, thereby demonstrating the highest predictive accuracy. In contrast, the GM(1,1) model produced the largest t-value. At a significance level of α=0.05, except for the monitoring results of the E5 sensor, the differences between the GM(1,1) model's predictions and the actual monitoring data were statistically significant. This suggests that the traditional grey model performed poorly in predicting slope stability during the stable phase.

B. EXPERIMENTS ON SLOPE BEFORE LANDSLIDE

In this section, the data of the five extensometer sensors (E1-E5) from April to May 2012 is selected (Preonzo slope has a landslide on May 15, 2012). Figure 7 (a)-(e) illustrates the experimental results obtained from these sensors during this critical period, and Table 3 illustrates the risk values corresponding to each sensor at that stage.

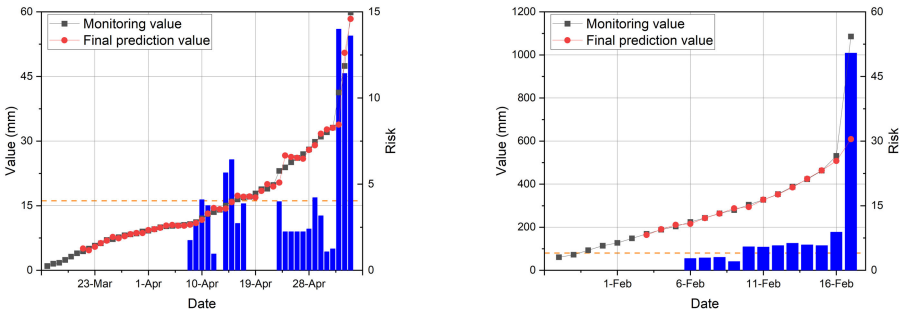
TABLE 3. Risk value of sensors before the landslide occurred on the Preonzo Slope.

Date	E1	E2	E3	E4	E5
23-Apr	-	184.21%	-	-	-
24-Apr	-	197.37%	-	-	-
25-Apr	-	205.33%	101.28%	-	-
26-Apr	-	164.27%	103.12%	-	-
27-Apr	-	193.86%	108.11%	-	-
28-Apr	-	186.81%	236.44%	-	-
29-Apr	192.42%	189.92%	118.41%	-	-
30-Apr	180.34%	203.68%	118.41%	-	-
1-May	169.54%	219.64%	141.86%	-	-
2-May	183.50%	254.02%	368.20%	-	-
3-May	236.79%	296.10%	386.69%	-	-
4-May	206.73%	254.18%	376.69%	-	-
5-May	145.15%	179.13%	255.64%	-	-
6-May	179.71%	224.55%	**416.85%**	197.16%	191.12%
7-May	**412.71%**	**498.61%**	880.34%	**419.80%**	**401.90%**
8-May	403.96%	484.05%	834.85%	294.40%	350.83%
9-May	246.33%	289.19%	466.63%	260.88%	184.51%
10-May	127.36%	148.36%	225.25%	200.36%	106.09%
11-May	130.54%	147.85%	321.98%	238.10%	145.18%
12-May	182.42%	206.78%	**421.27%**	250.14%	168.43%
13-May	212.43%	245.38%	478.98%	339.51%	213.31%
14-May	290.77%	334.74%	725.65%	**453.85%**	313.92%
15-May	**609.37%**	**765.68%**	780.64%	-	**610.32%**

TABLE 4. Results of T-test (before the landslide occurred on the Preonzo Slope).

Monitoring Point	16 Apr - 30 Apr	1 May - 15 May
E1	0.3408 (DF=14)	1.19567 (DF=14)
E2	0.4199 (DF=14)	1.17288 (DF=14)
E3	-0.24146 (DF=14)	1.31769 (DF=14)
E4	-0.70009 (DF=14)	1.01736 (DF=13)
E5	-0.01495 (DF=14)	1.31769 (DF=14)

Figure 7 (a)-(e) and Table 3 present the experimental results of five extensometer sensors monitored before the landslide on May 15, 2012. From the figures, it is evident that during the early monitoring period, the model's slope displacement predictions closely align with the actual displacement measurements, with minimal differences between the two. However, two days prior to the landslide event, a significant deviation appears, as the predicted values fall notably below the observed displacement values.



(a) Wolongsi landslide in 1971

(b) Longjingcun landslide in 2019

FIGURE 9. Experimental results on different landslides.

TABLE 5. Risk value before the Wolongsi landslide (1971).

Date	8-Apr	9-Apr	10-Apr	11-Apr	12-Apr	13-Apr	14-Apr	15-Apr	16-Apr	17-Apr
Risk	173.90%	272.00%	**409.86%**	376.63%	94.90%	-	**566.67%**	642.73%	272.42%	387.67%
Date	18-Apr	19-Apr	20-Apr	21-Apr	22-Apr	23-Apr	24-Apr	25-Apr	26-Apr	27-Apr
Risk	-	-	-	-	-	399.00%	224.64%	224.64%	224.64%	224.64%
Date	28-Apr	29-Apr	30-Apr	1-May	2-May	3-May	4-May	5-May		
Risk	240.13%	**422.45%**	319.45%	107.66%	124.66%	1400.24%	**1142.83%**	1360.65%		

TABLE 6. Risk value before the Longjingcun landslide (2019).

Date	6-Feb	7-Feb	8-Feb	9-Feb	10-Feb	11-Feb
Risk	275.80%	288.59%	302.22%	204.17%	**548.87%**	541.28%
Date	12-Feb	13-Feb	14-Feb	15-Feb	16-Feb	17-Feb
Risk	573.46%	628.05%	592.14%	**575.09%**	887.04%	5043.38%

From an early warning perspective, all sensors exceeded the risk threshold (400%) during the monitoring period, triggering standard warnings. Furthermore, the risk values remained above 100% for more than five days, prompting the sensors to issue advanced warnings. Following the issuance of advanced warnings, the risk values for all sensors surpassed the critical risk threshold, leading to the generation of highest-level warnings. An exception was observed with the E3 sensor, which issued a standard warning on May 6 and an advanced warning on May 11. The remaining four sensors followed a similar timeline, generating standard warnings on May 7 and advanced warnings on May 12. The landslide on the Preonzo slope ultimately occurred on May 15.

The dynamic prediction model proposed in this paper effectively forecasted slope displacement and provided timely early warnings of potential instability in the days leading up to the landslide event.

To assess the accuracy of the prediction model proposed in this paper, we conducted comparative experiments involving the proposed model, the standard GM(1,1) model, and the GM(1,1) model with a window size of 5. Figure 8 illustrates the comparative results for the pre-landslide phase of the Preonzo slope (April to May 2012). As depicted in the figure, the displacement predictions of the traditional GM(1,1) model showed poor alignment with the actual displacement values, highlighting its limited predictive performance. In contrast, both the GM(1,1) model with a window size of 5 and the proposed model generated predictions consistent with the actual displacement values. Among these, the proposed model demonstrated the smallest prediction errors, indicating superior accuracy.

To further evaluate the performance of the proposed model during the pre-landslide phase, we performed T-tests for two distinct sub-phases: April 16-30 and May 1-15. Table 4 presents the T-test results, revealing that the t-value for the final 15 days was significantly higher than that for the initial 15 days. This discrepancy is attributed to the sharp increase in actual slope displacement immediately preceding the landslide, which surpassed the predicted values. These deviations notably elevate the corresponding Risk value, thereby establishing a robust basis for issuing timely landslide warnings.

C. EXPERIMENTS ON DIFFERENT LANDSLIDES

To further assess the model's applicability across different scenarios, two additional landslide datasets (the Wolongsi landslide and the Longjingcun landslide) were used to evaluate its early warning performance.

The Wolongsi landslide, a shallow loess landslide, occurred at 3:00 AM on May 5, 1971, in Jintai District, Baoji

TABLE 7. Comparisons between different models.

	Our Model	Machine learning-based models	Other grey system-based prediction models
Dynamic	yes	yes/no	no
Data Requirements	small size, easy to obtain	big size, difficult to obtain	small size, easy to obtain
Resource Requirements	less	more	less
Computational Efficiency	high	lower than other two models	high
Accuracy	good	good	lower than other two models

City, Shaanxi Province. Figure 9(a) and Table 5 show the early warning results for the slope prior to the landslide using the proposed model. The figure reveals that, during the initial monitoring period, the model's slope displacement predictions closely matched the actual displacement measurements, with minimal differences. However, starting on April 23, the differences began to increase, and a significant deviation emerged two days before the landslide, with the predicted values falling notably below the observed displacement.

From an early warning perspective, the monitoring values first exceeded the risk threshold (400%) on April 10, triggering a normal warning. This warning was lifted three days later. However, on April 14, the monitoring values again surpassed the risk threshold, reissuing a normal warning that lasted for four days without escalating to an advanced warning. On April 29, the monitoring values exceeded the risk threshold for the third time. An advanced warning was issued on May 3, followed by the highest-level warning on May 4, just before the landslide occurred.

Longjingcun landslide is located in Xingyi City in the southwestern of Guizhou Province, which occurred on February 17, 2019. Figure 9(b) and Table 6 show the early warning results for the slope prior to the landslide using the proposed model. From the figure, it is evident that during the early monitoring period, the model's slope displacement predictions closely align with the actual displacement measurements, with minimal differences between the two. However, two days prior to the landslide event, a significant deviation appears, as the predicted values fall notably below the observed displacement values.

The risk value began calculated from February 6, and continues until the landslide occurred. The monitoring values first exceeded the risk threshold (400%) on February 10, triggering a normal warning. An advanced warning was issued on February 14, followed by the highest-level warning on February 15, which is 2 days before the landslide.

These results demonstrate that the proposed model effectively and efficiently issues early warnings in various landslide scenarios.

V. DISCUSSION

A. COMPARISONS BETWEEN DIFFERENT MODELS

This paper introduces a dynamic landslide warning model that integrates the concept of risk with grey system theory, utilizing a minimal amount of real-time monitoring data. Table 7 compares the proposed model with machine learning-based prediction models and other grey system-based prediction models.

In terms of adaptability, the proposed model surpasses other grey system-based models by leveraging the most recent monitoring data, offering superior adaptability and dynamism. For machine learning-based models, adaptability depends largely on the flexibility of the underlying machine learning algorithm.

Regarding data and resource requirements, the proposed model relies on monitoring a single data dimension (displacement) and does not require extensive historical datasets or additional data types. While the GM(1,1) model uses similar data inputs, its lack of a mechanism to prioritize the latest data results in slightly higher resource consumption compared to the proposed model. Nevertheless, the resource demands of the proposed model remain significantly lower than those of machine learning-based models.

In terms of computational efficiency, assuming there are n time series data points, the main computational complexity of the traditional GM(1,1) model stems from matrix operations, with a time complexity of $O(n^2)$. The proposed model, employing a sliding window approach to use the latest k monitoring data, achieves a time complexity of $O(k^2)$. In contrast, machine learning-based models often exhibit a time complexity far exceeding $O(n^2)$. The proposed model's rapid computation, taking less than one second per prediction in practical applications where sensors typically collect data at intervals of no less than five minutes, makes it highly efficient and well-suited for real-world scenarios.

In terms of accuracy, experimental results demonstrate the proposed model's strong performance in landslide prediction across various time periods and locations, providing timely warnings. T-test results further reveal that the proposed model achieves the smallest t-value compared to other models, such as the traditional GM(1,1) model, indicating the highest consistency between predictions and observed data. This underscores the model's high accuracy.

Previous studies have shown that GM(1,1)-based prediction models can effectively forecast certain specific slopes but tend to exhibit the highest mean absolute error, reflecting relatively high but less consistent accuracy. Machine learning-based models, which often rely on large datasets for training or integrate multiple models, also achieve high accuracy.

However, because the proposed model relies solely on a single parameter (slope displacement) for predictions, in certain extreme cases (such as the failure of specific sensors

measuring slope displacement), the proposed prediction model may become ineffective. In such scenarios, machine learning-based models, which leverage multi-dimensional data, may offer better performance. Future enhancements to address this limitation are discussed in detail in Section V-B.

B. FUTURE DIRECTIONS

Further research could focus on the following areas:

(1) Enhancing the Precision of Grey Models

While this study proposes a dynamic landslide warning model based on the traditional GM (1,1) model, there is room for improvement in refining the GM (1,1) model itself. These refinements could lead to more accurate predictive results and greater reliability in practical applications.

(2) Improving the Accuracy of Monitoring Data

Sensor-acquired data is often subject to errors that can compromise predictive accuracy. Traditional sensor technologies may produce imprecise data, which reduces the effectiveness of predictive models. However, advancements in Global Navigation Satellite System (GNSS) technology and image processing have made it possible to collect real-time dynamic slope data with greater accuracy and reliability. These improvements significantly enhance the overall precision of predictions.

(3) Automating the Frequency of Data Monitoring

Based on real-time monitoring data, we propose the dynamic prediction early warning model. The model's accuracy largely depends on the quality of the monitoring data, and the sliding window mechanism ensures that the most up-to-date data is used for risk assessment. The model can be dynamically updated with real-time data. Future research could explore mechanisms to adjust the data monitoring frequency based on specific conditions (e.g., increasing the frequency of data monitoring during heavy rainfall) to further support the model's dynamic updates. This would ensure the model's continued effectiveness over time.

(4) Implementing Self-Supervised Mechanisms

The prediction model proposed in this paper relies heavily on real-time monitoring data. While this approach ensures real-time responsiveness and dynamic adaptability, it is highly sensitive to the quality of the data. Introducing a self-supervised mechanism, similar to the approach described in Reference [31], could help address issues arising from unreliable or noisy data. Such mechanisms could enhance data quality, enabling more effective prediction and early warning.

(5) Automating Threshold Determination

The dynamic landslide warning model in this study requires manually specified thresholds for the warning level (R_T) and error calculation (R_C). Using big data technologies (such as using the data of rainfall amount, rate, and duration to calculate parameters automatically), these parameters could be automatically estimated based on data analysis, reducing the risk of inaccuracies caused by manual parameter settings. Automating this process would improve the model's reliability and applicability in diverse scenarios.

(6) Improving the Robustness of the Prediction Model

Although the proposed model demonstrates effective performance in various landslide prediction scenarios, it may become ineffective in certain extreme cases, such as the failure of specific sensors monitoring slope displacement. To enhance its applicability across diverse geographical and environmental contexts, it is essential to improve the model's robustness. One approach is to deploy multiple sensors on slopes at risk of landslides while optimizing sensor placement to minimize the number of sensors required. Another approach involves employing machine learning techniques to filter out invalid data and generate accurate replacements, thereby improving the reliability of the input data. Lastly, adopting integrated prediction models can further enhance accuracy by leveraging the strengths of different models to produce more reliable and robust predictions.

VI. CONCLUSION

This paper proposes a dynamic landslide early warning model based on grey system theory. The proposed model combines the GM(1,1) model with the concept of a sliding window to ensure predictions are based on the most recent data, thereby minimizing computational errors associated with outdated historical data. Additionally, a three-level early warning mechanism is designed, leveraging prediction results and real-time monitoring data to establish a robust and comprehensive landslide early warning framework. Experimental results validate the model's effectiveness in accurately predicting landslides and issuing timely warnings.

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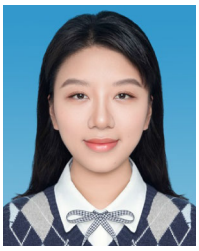

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